

SQL Stored Procedures

A **stored procedure** is a group of SQL statements stored in a database that can be executed whenever required.

1. Basic Procedure

Example: Display all employee details.

```
DELIMITER //
```

```
CREATE PROCEDURE GetEmployees()
```

```
BEGIN
```

```
    SELECT * FROM Employee;
```

```
END //
```

```
DELIMITER ;
```

```
CALL GetEmployees();
```

Explanation:

- CREATE PROCEDURE – Creates a procedure.
- BEGIN and END – Define the procedure body.
- CALL – Executes the procedure.

2. Procedure for INSERT

Example: Insert employee details into the table.

```
DELIMITER //
```

```
CREATE PROCEDURE InsertEmployee(
```

```
    IN id INT,
```

```
    IN name VARCHAR(50),
```

```
    IN salary INT
```

```
)  
  
BEGIN  
  
    INSERT INTO Employee  
  
    VALUES (id, name, salary);  
  
END //
```

DELIMITER ;

```
CALL InsertEmployee(1, 'Rahul', 25000);
```

Explanation:

- IN passes values to the procedure.
 - INSERT INTO adds a new record.
 - CALL executes the procedure with the given values.
-

3. Procedure for UPDATE

Example: Update an employee's salary.

DELIMITER //

```
CREATE PROCEDURE UpdateEmployee(  
  
    IN id INT,  
  
    IN new_salary INT  
  
)  
  
BEGIN  
  
    UPDATE Employee  
  
    SET salary = new_salary  
  
    WHERE emp_id = id;  
  
END //
```

DELIMITER ;

CALL UpdateEmployee(1, 30000);

Explanation:

- UPDATE modifies existing records.
- SET specifies the new value.
- WHERE identifies the employee to update.

Note: Here, emp_id is the employee ID column.

4. Procedure for DELETE

Example: Delete an employee using their ID.

DELIMITER //

```
CREATE PROCEDURE DeleteEmployee(  
    IN id INT  
)  
BEGIN  
    DELETE FROM Employee  
    WHERE emp_id = id;  
END //
```

DELIMITER ;

CALL DeleteEmployee(1);

Explanation:

- DELETE FROM removes a record.
- WHERE specifies which record to delete.
- CALL executes the procedure.

